

C.3.3 Fluctuations des distributions de rapidité

On a vu que

$$2\pi \rho_s(\theta) = 1_{[\nu]}^{\text{dr}}(\theta) = \int d\lambda \left(\delta(\theta - \lambda) + \nu(\theta) C(\theta, \lambda) \right), \quad (\text{C.18})$$

d'où

$$2\pi \delta\rho_s(\theta) = \int d\lambda C(\theta, \lambda) \rho_s(\lambda) \delta\nu(\lambda). \quad (\text{C.19})$$

Ainsi,

$$\delta\rho(\theta) = \rho_s(\theta) \delta\nu(\theta) + \nu(\theta) \delta\rho_s(\theta) \quad (\text{C.20})$$

$$= \int d\lambda \delta(\theta - \lambda) \rho_s(\lambda) \delta\nu(\lambda) + \int d\lambda \nu(\theta) C(\theta, \lambda) \rho_s(\lambda) \delta\nu(\lambda), \quad (\text{C.21})$$

$$= \int d\lambda \left(\delta(\theta - \lambda) + \nu(\theta) C(\theta, \lambda) \right) \rho_s(\lambda) \delta\nu(\lambda). \quad (\text{C.22})$$

En réinjectant (C.11) et (C.22) dans la définition

$$\langle \delta\rho(\theta') \delta\rho(\theta) \rangle_w = - \left[L \frac{\delta^2(\mathcal{S}_{YY} - \mathcal{W})}{\delta\rho \delta\rho} \right]^{-1}(\theta, \theta'), \quad (\text{C.23})$$

on obtient finalement

$$\langle \delta\rho(\theta') \delta\rho(\theta) \rangle_w = -\frac{1}{L} \int d\lambda \frac{\left(\delta(\theta - \lambda) + \nu(\theta) C(\theta, \lambda) \right) \left(\delta(\theta' - \lambda) + \nu(\theta') C(\theta', \lambda) \right)}{s''(\nu(\lambda))} \rho_s(\lambda) \quad (\text{C.24})$$

$$= \frac{1}{L} \mathcal{D}^{-1}(\theta, \theta') + \frac{1}{L} \mathcal{B}(\theta, \theta'), \quad (\text{C.25})$$

où

$$\mathcal{D}^{-1}(\theta, \theta') = (\rho_s(\theta) \nu(\theta) (1 - \nu(\theta))) \delta(\theta - \theta'), \quad (\text{C.26})$$

$$\mathcal{B}(\theta, \theta') = \left[\nu(\theta') \rho_s(\theta) \nu(\theta) (1 - \nu(\theta)) + \nu(\theta) \rho_s(\theta') \nu(\theta') (1 - \nu(\theta')) \right] C(\theta, \theta') \quad (\text{C.27})$$

$$+ \nu(\theta) \nu(\theta') \int d\lambda \rho_s(\lambda) \nu(\lambda) (1 - \nu(\lambda)) C(\theta, \lambda) C(\theta', \lambda). \quad (\text{C.28})$$

Cette expression coïncide numériquement avec le résultat obtenu en (4.59).

